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22 April 1960
SUPERSEDING
NAVORD OS 7190 B
24 February 1958

MILITARY SPECIFICATION

HOUSING, BATTERY, MARK 1 MOD 0

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the Housing, Battery, Mark 1 Mod 0, LD 479302. This battery housing is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Fiber

MILITARY

MIL-P-116	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 479302	Housing, Battery, Mark 1 Mod 0

and all documents listed therein.

FSC 1105

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PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400	General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28 and 1950 Supplement	Screw Thread Standards for the Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

3. REQUIREMENTS

3.1 Description. - The Battery Housing Mk 1 Mod 0 is an aluminum alloy watertight container which is used for retention and protection of the high voltage battery.

3.2 General requirements.

3.2.1 Pre-Production samples. - Unless otherwise specified in the contract or order, pre-production samples of the Battery Housing Mk 1 Mod 0 shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. Provision of the sample shall be in accordance with 4.1.2.

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3.2.2 Materials. - Unless otherwise specified in the contract, requisition or order all materials and purchased parts entering into the final assembly of the Battery Housing Mk 1 Mod 0 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUORD LD 479302.

3.2.3 Finishes, coatings, constructions and dimensions. - Finishes, coatings, construction, and dimensions for the Battery Housing Mk 1 Mod 0 shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUORD LD 479302.

3.2.4 Marking. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in BUORD LD 479302.

3.2.5 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings will be made without prior revisions and correction of the applicable drawings.
See 3.2.9.

3.2.6 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility, in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and Supplements thereto. The class of fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the supplement to the aforementioned Handbook H28.

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3.2.8 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.2.9 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions. - For the purpose of this specification, the following definitions are applicable.

3.3.1 Battery housing. - Reference to battery housing unless otherwise indicated, shall mean Housing, Battery Mk 1 Mod 0 as covered by BUORD LD 479302.

3.4 Performance requirements and product characteristics. - The battery housing shall meet the following performance requirements and product characteristics:

3.4.1 Leakage. - The battery housing (dwg. 1467793) shall be capable of withstanding without leakage, an external hydrostatic pressure of fifteen (15) pounds per square inch when tested in accordance with method described in paragraph 4.5.4.1.

3.5 Pre-production and periodic production requirements.

3.5.1 Hydrostatic strength. - The battery housing (dwg. 1467793) must withstand an external hydrostatic pressure of 1000 psi without leakage or permanent deformation in excess of that specified in 4.6.1 when tested in accordance with method described in 4.6.1. After meeting these requirements the battery housing must meet the performance requirements and product characteristics of this specification.

3.5.2 Salt fog. - The battery housing must meet the requirements of 3.5.1 after being subjected to the salt fog test described in 4.6.2.

3.6 Clean-up. - Prior to and after final assembly all parts, components and the assembly shall be thoroughly cleaned of metal chips and other foreign matter. Burrs and sharp edges shall be removed, prior to anodizing.

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3.7 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of marking of parts and assemblies, anodizing, painting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of three (3) battery housings shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for pre-production tests at an activity designated in the contract or by the Bureau of Ordnance, for purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Lot size. - As applied to Government inspection of units of product the term "lot" shall mean "Inspection lot", i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be as determined by the Government Inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment or other purpose.

4.1.4 Periodic production samples. - From each succeeding lot of one hundred (100) or less, a random sample of two (2) battery housings

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will be selected by the Government inspector for submission to a Bureau of Ordnance designated activity to determine compliance of the samples with the requirements of the contract, specification and drawings. The battery housing shall be considered satisfactory only if each unit meets all specification and pertinent drawing requirements, the performance requirements and product characteristics of 3.4 and the environmental requirements of 3.5.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of Tests. - The inspection and testing of the battery housing shall be classified as follows:

4.3.1 Acceptance tests. - These tests are to be accomplished on battery housings being submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production. Acceptance tests shall either be performed by the manufacturer and witnessed by the Government inspector, or shall be conducted by the Government inspector. These tests are defined in 4.5.

4.3.2 Pre-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.4 which are representative of the production of the battery housing after the award of the contract to determine that the lot and production meet the requirements of this specification. These tests are detailed in 4.6. Acceptance shall be based on no defects in a sample. Failure of a sample to comply with these requirements will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.4.1 Tank of sufficient size to contain the battery housing with pressure cap attached and capable of being pressurized to 15 ± 3 psi.

4.4.2 A steel pressure cap as shown in figure 1.

4.4.3 "0" ring (33-P-1561-1000).

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4.4.4 Hydrostatic pressure gage, reading 0-20 psi, accurate to ± 1 psi.

4.5 Acceptance tests.

4.5.1 The Contractor or Government inspector shall conduct acceptance tests as specified in 4.3.1 to assure compliance with the requirements of 3.2 and 3.4.1.

4.5.2 Test conditions. - Unless otherwise specified the test procedure of 4.5.4 shall be accomplished under the following conditions.

4.5.2.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35.0 degrees C).

4.5.2.2 Altitude: Normal ground.

4.5.2.3 Vibration: None.

4.5.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.5.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.4 Test procedure.

4.5.4.1 Leakage test. - This test shall be performed prior to installation of screws (1490092) and (1476493-19). The battery housing (1467793) with "0" ring (4.4.3) installed, and with pressure cap (4.4.2) held in place by C-clamps as shown in Figure I, shall be placed in the tank (4.4.1) and completely immersed in water. Increasing hydrostatic pressure shall then be applied to the outside of the housing until a pressure of 15 ± 3 psi, as determined by means of 4.4.4, is reached. The pressure shall be maintained for a minimum period of ten (10) minutes. After housing has been removed from the tank, and external residual water has been cleaned off, the inside of the housing shall be examined for visible evidence of leakage. If no leakage is detected the housing shall be considered to have met the requirements of this test. (NOTE: "0" ring (4.4.3) shall be replaced with original "0" ring after completion of this test.)"

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4.6 Pre-production and periodic production tests.

4.6.1 Hydrostatic strength test. - This test shall be performed prior to the installation of screws (1490092) and (1476493-19). Prior to assembly of battery housing (1467793), dimensions "X" and "Y" shown in Figure 1 shall be measured to an accuracy of one thousandth (.001) of an inch and reading recorded. The battery housing with "0" ring (4.4.3) installed, and with pressure cap (4.4.2) held in place with C clamps as shown in Figure 1 shall then be placed in a suitable pressure pot and completely immersed in water. Apply an external hydrostatic pressure of 1000 ± 20 psi to the battery housing and maintain for a period of not less than ten (10) minutes. Remove assembly from the pressure pot. After external residual water has been cleaned off remove pressure cap and examine inside of the housing for visible evidence of leakage. Remeasure dimensions "X" and "Y" at the same locations previously measured. If there is no visible evidence of leakage, and if there is reproducibility of the measurements within two thousandths (.002) of an inch, the housing shall be considered to have met the test requirements. (NOTE: "0" ring (4.4.3) shall be replaced with original "0" ring after completion of this test).

4.6.2 Salt fog test. - Subject the battery housing, with mounting cover assembly (1519821) installed, to salt fog atmosphere for a period of 96 hours ± 4 hours as follows: Suspend the battery housing in a suitable salt fog corrosion cabinet by means of a waxed cord or plastic thread in such a position that it will be most vulnerable to the salt fog, and inclined at an angle of 15° to prevent accumulation of condensate on horizontal surfaces. Temperature in the cabinet shall be maintained at 95° ± 3°F and the relative humidity at 87 ± 3%. The salt fog atmosphere shall be maintained by passing air at 12 psi through a water tank maintained at 110°F and passing the saturated air through an aspirator which draws a 20 ± 2% salt solution, having a pH between 6.5 and 7.2, into the air stream which passes through a small nozzle into the cabinet. The test shall be continuous except for a daily interruption to inspect the battery housing and to change the condensate collectors. Concentration and pH measurements shall be made on the condensate samples. Battery housings meeting the requirements of 3.5.1 after completion of this test shall be considered to have met the specified requirements.

4.7 Acceptance criteria. - Battery housings which fail to meet any of the performance requirements, product characteristics, or environmental requirements of this specification shall be regarded as defective. If the number of defective units exceeds the number of defectives

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permitted by the associated acceptance criteria, the lot represented will be rejected. Battery housings meeting all the performance requirements, product characteristics and environmental requirements of this specification shall be stocked for service use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning. - Unless otherwise specified, the Battery Housing shall be cleaned in accordance with Method C-1 of Specification MIL-P-116.

5.1.1.2 Drying. - Unless otherwise specified the Battery Housing shall be dried in accordance with one or more of the procedures given in Specification MIL-P-116 provided the procedure is not injurious to the item.

5.1.1.3 Preservative application. - Not applicable.

5.1.1.4 Unit packaging. - Unit packaging of the Battery Housing shall be in accordance with Method III of Specification MIL-P-116.

5.1.2 Level B. - Not applicable.

5.1.3 Level C. - The Battery Housing packaged in accordance with 5.1 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. - In addition to the markings required by contract or order, unit packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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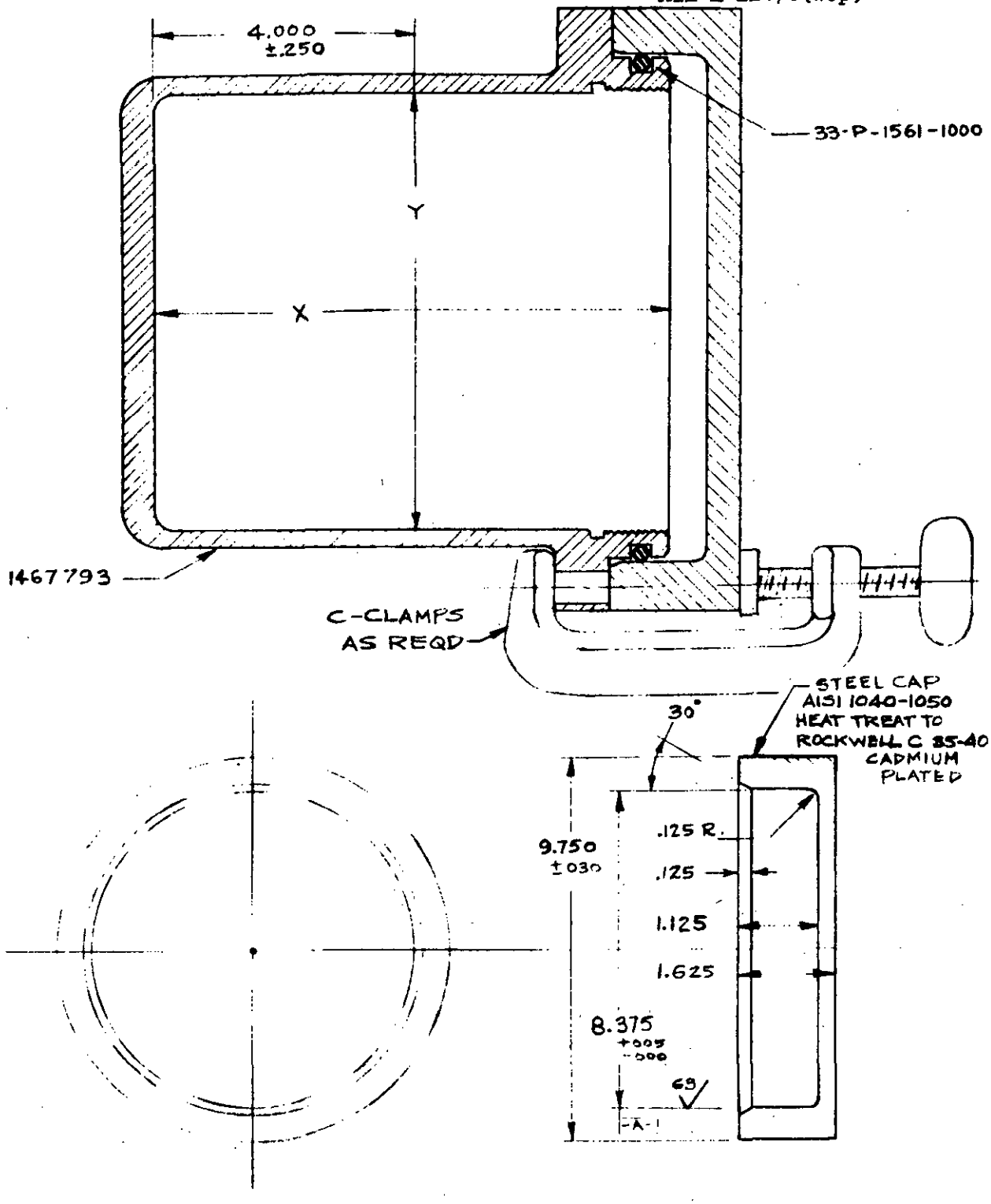


FIGURE 1

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